VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. V Semester

(22PE1AM302) EXPLAINABLE AI

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the concepts of explainable AI and interpretable machine learning
- To understand concepts of black-box machine learning, explainable AI methods
- To analyse various explainable AI methods
- To understand various applications of explainable deep learning
- To identify various challenges related to explainable AI

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the context of the machine learning applications with respect to explain ability and interpretability

CO-2: Analyze concepts of various ML model validation, evaluation, and performance visualization for both supervised and unsupervised learning

CO-3: Apply state of the art explainable AI tools, techniques and interpretable methods for various AI applications

CO-4: Corelate explainable deep learning techniques and its applications

CO-5: Interpret various XAI applications and future challenges related to Explainable AI

COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	2	2	2	1	-	-	1	-	-	-	-	2	-	1
CO-2	2	-	2	1	2	-	-	2	-	-	-	-	2	2	2
CO-3	-	2	3	2	2	-	-	1	-	-	-	-	2	-	2
CO-4	-	-	-	1	-	-	-	2	-	-	-	-	2	-	2
CO-5	2	-	3	1	1	-	-	1	-	-	-	-	2	-	2

UNIT-I:

Introduction: Black-Box problem, Goals, Brief History, Purpose, Societal Impact, Types of Explanations, Trade-offs, Taxonomy, Flowchart for Interpretable and Explainable Techniques.

Pre-model Interpretability and Explainability: Data Science Process and EDA, Exploratory Data Analysis, Feature Engineering.

UNIT-II:

Model Visualization Techniques and Traditional Interpretable Algorithms: Model Validation, Evaluation, and Hyperparameters, Model Selection and Visualization, Classification Model Visualization, Regression Model Visualization, Clustering Model

Visualization, Interpretable Machine Learning Properties, Traditional Interpretable Algorithms

UNIT-III:

Model Interpretability: Advances in Interpretable Machine Learning: Interpretable vs. Explainable. Algorithms, Tools and Libraries, Ensemble-Based, Decision Tree-Based, Rule-Based Techniques, Scoring System.

Post-Hoc Interpretability and Explanations: Tools and Libraries, Visual Explanation, Feature Importance, Example-Based.

UNIT-IV:

Explainable Deep Learning: Applications, Tools and Libraries, Intrinsic, Perturbation, Gradient / Backpropagation.

UNIT-V:

Explainability: Time Series Forecasting, Natural Language Processing, and Computer Vision XAI: Challenges: Properties of Explanation, Categories of Explanation, Taxonomy of Explanation Evaluation.

XAI: Future: Formalization of Explanation Techniques and Evaluations, Adoption of Interpretable Techniques, Human-Machine Collaboration, Collective Intelligence from Multiple Disciplines, Responsible AI (RAI), XAI and Security, Causality and XAI.

TEXT BOOK:

1. Explainable Artificial Intelligence: An Introduction to Interpretable Machine Learning, Uday Kamath and John Liu, 1st Edition, Springer, 2021

REFERENCES:

1. Explainable AI with Python, Leonida Gianfagna and Antonio Di Cecco, 1st Edition, Springer International, 2021
2. Hands-On Explainable AI (XAI) with Python, Denis Rothman, 1st Edition, Packt Publishing, 2020